A

Project Report

On

VoiceAuth (Detection of Fake Voices)

Submitted in Partial Fulfillment of the Requirements

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BACHELOR OF TECHNOLOGY

in

Computer Science and Design.

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DECLARATION

We hereby declare that the project work presented in this report entitled **"VoiceAuth-Detection of Fake Voices"**, in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Galgotias College of Engineering & Technology, Greater Noida, Uttar Pradesh, submitted to Dr. A.P.J. Abdul Kalam Technical University, Uttar Pradesh, Lucknow is based on our own work carried out at **Department of Artificial Intelligence & Machine Learning**, Greater Noida. The work contained in the report is true and original to the best of our knowledge and project work reported in this report has not been submitted by us for award of any other degree or diploma.

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CERTIFICATE

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ABSTRACT

The increasing level of realism with artificially-generated voice that is unable to be differentiated from authentic human speech poses a severe concern in both the areas of security and social integrity [11]. Voice biometrics are predicated on the assumption that the system is identifying who spoke when, and only verifies that the person is the one speaking; thus the system does not establish the authenticity or liveliness of the audio signal that it was identifying. Since voice biometric services are designed to identify who spoke when based on voiceprints, and there is no mechanism to verify the authenticity of the audio in terms of being a synthetic or altered recording, audio that is generated through sophisticated audio synthesis systems can be used to circumvent established security systems.

Therefore, the threat of widespread digital impersonation (i.e., phishing) and financial fraud will persist. In response to this gap, this project aims to create a sound methodology for accurately identifying synthetic voices. The project begins by developing a comprehensive database with a sufficiently large number of synthentic and bonafide audio samples that have been validated by third party sources. Feature engineering for this audio data will utilize advanced acoustic features such as MelFrequency Cepstral Coefficients (MFCCs) and Constant Q Cepstral Coefficients (CQCCs). These features will allow for the identification of the very subtle and measurable differences that exist between the voice of a human, and one that is artificially produced. These derived features will be used to train and evaluate various machine learning classifiers, including Random Forest, Support Vector Machines (SVM), and a light form of Convolutional Neural Network (CNN).The final product is a dual layer of authentication. Authentication will require both an authenticity verification and confirmation, as well as a verification of liveliness of the voice that has been received and recorded.

Keywords

MFCC (Mel-Frequency Cepstral Coefficients), Chroma Features, Spectral Features, Random Forest Classifier, Support Vector Machine (SVM), Convolutional Neural Network (CNN).

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CHAPTER 1

INTRODUCTION

Voice Synthesis has now entered the new age of our world through Deep Learning. With the advent of Deep Learning, it is now possible to synthesize speech with an incredibly high degree of acoustic accuracy (i.e., how closely the synthetic voice sounds like the real-life speech, how well the synthetic voice can reproduce the phonetic aspects of a word, the timing and emotion behind the words, etc.) which allows for many of the additional nuances of speech to be included in the synthesis of a voice. Deep Learning has enabled the multitude of applications for voice technology that are currently being utilized by businesses in a variety of industries. For example, voice technology can produce narrated audio for media productions, as well as allow for localization of the audio being produced (i.e., the production of voice-overs). However, perhaps, the most important utilization of today's voice technology has been to develop a synthetic voice for an individual with a disability that prevents them from producing their own voice by providing a means to support who they are as an individual while recreating the personal characteristics associated with their speech [17].

1.1 FAKE VOICE DETECTION

Unfortunately, the technology and more broadly the internet has become a sophisticated environment (Digital and Commercial) conducive to the establishment of increased numbers of encroachments on Privacy and Security. The Rate at which this increase is occurring has far outpaced the ability of businesses and consumers to respond in any meaningful way. One significant issue resulting from this unchecked technological advancement is the introduction of new possibilities surrounding the generation of artificial speech using Machine Generated Audio ("Deepfake") technologies. This creates a scenario in which human listeners cannot distinguish between authentic versus artificial sounds or voices. At the same time, these technologies create new vulnerabilities in terms of verifying identity, as well as creating opportunities for the creation of Artificially Generated audiograms/ presentations that can be utilized without detection. The primary objective of this research is to address a long-standing limitation with existing Voice Identification (VI) and Automatic Speaker Verification (ASV) technologies. Moreover, existing VI and ASV technologies rely heavily on matching static features associated with a given user instead of focusing on the verification of whether an audio signal being presented for verification is live and emanated from a specific source.

As such, in addition to developing a method which will allow a user to detect and remove audio that has been produced illegally, we shall also develop technology that will identify audio that has been illegally produced. The primary purpose of this research will be to provide a method for validating acoustic authenticity of real time audio.

The subsequent step of analysis will be an in-depth feature extraction phase. This phase will fundamentally differ from the previously discussed methods. The previously discussed methods examined only acoustic data collected from the surface of speakers through the use of signal data collected at the surface of speakers. The features that this project will use will be based on the analysis of acoustics in order to determine what has been manufactured based on the manner it

was manufactured instead of being collected from the surface of speakers.

Example features that will be extracted for this project's assessment include multi-cepstral coefficients, etc., which will show very small variances in the acoustic signal as a result of the method of manufacture,a process that is carried out by a machine that does not have the capabilities to produce the complex spontaneous speech waveforms that are obtained from the natural voice that is sampled. The team is in the process of creating solutions to the rising number of challenges in identifying and differentiating fraudulent, artificial, and other types of synthetic voice technology.The team believes that, through the use of advanced technologies, they will be able to provide users with the means to assess, thus being able to confidently engage in the use of voice technology, to the extent that all, including the general public, business entities, and voice technology service providers, will be able to confidently engage in an orderly fashion in the use of voice technologies.

The team believes that fraud will occur at its most harmful level when either the person being defrauded, or when someone impersonating the victim through AI technology, has already been defrauded. Thus, the team sees one of the most intriguing aspects of AI technology as finding ways to produce more sophisticated voice applications that will promote greater user confidence.

Risks associated with unregulated deep fake audio are no longer mere speculation, and are quickly becoming very real, with significant potential to threaten the overall economic stability, business security, and people's confidence in their individual rights. Deception using deep fakes is particularly acute in areas of the bank's voice command-based (or voice-driven) transaction systems; and in political contexts, being able to verify the true source of a voice is critical to verifying the accuracy of the information being reviewed. With the general public and a growing number of businesses depending on voice command technologies to transact business real time in an electronic medium, the use of deep fake audio to deceive someone is becoming ever more likely [16]. As society continues to adopt digital voice technologies at a high rate, it has increased the likelihood that users are unable to confirm or validate the physical origin of their voice signal. Because of this, there is a systemic risk associated with the inability to verify voice; therefore, countermeasures must be developed to eliminate the risk. This is the objective of the project.

1.2 MOTIVATION

The impetus for this investigation stems from the advancement of technology to create an exact replica of someone's voice using artificial intelligence—such as the ability to create nearly identical copies of people’s voices—is now available to anyone who has enough audio from that person to use as a basis. This new technology allows for the ability to produce highly realistic replicas of human voices easily, and it is no longer restricted to the traditional research and development community. These developments provide new creative and therapeutic possibilities (e.g., developing personal digital assistants, assisting individuals who suffer from communication disabilities). However, with the widespread availability of this new technology comes an unprecedently large, complex security problem for people who conduct business or otherwise communicate using voice signatures. We are currently witnessing a rapid decline in the confidence and use of voice biometric authentication and verification due to the technological advances developed to protect voice biometric authentication and verification.

Another worry is that around the world people are relying more and more on voice-activated devices (like Siri, Alexa, etc.) for some of the most sensitive and security-related businesses in the world (e.g., banking, telecommunications, and large corporations) as a way to replace traditional password systems with voice authentication as a primary method of accessing sensitive information or performing high-value transactions. Criminals use this heavy reliance on voice authentication as an easy way to make money by attacking the system. The most basic point of vulnerability comes from the fact that many of these voice verification systems use only one factor in their security design.

Because of this gap in the current voice biometric authentication system, sophisticated, high-fidelity synthetic audio known as deepfake allows for easy access to the identity verification process, thereby making the entire voice biometric authentication system a single point of failure and creating a significant opportunity for financial fraud and identity theft to occur.

The security vulnerabilities from the aforementioned malicious sources and others affect both the person who is affected as well as the methods in which that person is connected to the digital world. The reality is that since automatic digital commerce has already been established in commercial markets, with a continuous rise in risks associated with usage of such services, individuals are now at an increased risk of having their financial data compromised due to automated payment mechanisms and having their identity stolen, thus the perceived risks associated with using digital commerce will be greater than ever, and therefore people will be less willing or able to utilize the many digital modalities of interacting with commerce. Corporations will also be at a disadvantage due to the significant amount of asset value and reputation that they will potentially lose as a result of the inherent vulnerability of voice-activated communications; with the risk of any voice-based trade currently being created or transferred via voice "synthesis" processing means increasing daily as the result of the numerous instances of successful "deepfake" attack. The bottom line is the integrity of voice communication (from both call center and government communications) has no sustainable future due to the ever-growing number of threats associated with these types of centralized and/or automatic communications and without effective, timeliness, real-time, protective means, this trend will continue to drive systemic risks toward the use of these abusive centralized means and ultimately create ongoing losses.

There are two key differences between synthetic and natural speech; how each sounds are produced, and how they are created. A major point of difference between how both types of speech are created is that synthetic speech produces sounds consistently and uniformly; therefore there won't be small variations in how sounds are made (i.e., jet noise...i.e., a person saying "hi"), and in addition, no type of sound will produce varying pitch, timing, or duration.

Technological advances in synthesizing near identical (or indistinguishable) sounds make it impossible for any waveforms (i.e., the voice), to be identical (or even nearly identical); and because rendering synthetic sounds entails the application of post-processing effects to each audio/speech signal prior to distributing it to the end user, every individual piece of audio (i.e., speech) will not be produced in a manner to produce the exact same piece of audio (i.e., speech), regardless of how similar or different the audio (speech) pieces sound. The mathematical probability of each audio (speech) segment produced to be rendered/reproduced identically is equal to zero % (0%).

Some of the additional processing steps can distort or create additional distortion of sound sources being produced via the process of synthesised human speech. While the majority of these distortions are not perceptible by the average person, when viewed closely or in a quiet environment the distortion will be clearly visible, thus providing our detection system with the ability to detect the vast majority of all types of synthetic speech patterns produced by the rapidly advancing technologies used today for producing lifelike, synthetic, humanly generated speech waveforms.

CHAPTER 2

LITERATURE REVIEW

As the technology for creating synthetic audio improves rapidly, it becomes more important to research how people can be identified as original speakers of audiobooks. Not only has artificial intelligence made significant advancements in AI-generated (synthetic sound) that replicates human-sounding voices, but other fields including providing assistance for people with disabilities, creating virtual assistants (intelligent) using AI technology, and developing new types of electronic books are greatly benefitting from AI-generated (synthetic speech).

The dangers posed by AI-generated (synthetic sound) are substantial. By utilising AI (to create sound), identity fraud can occur through AI-generated audio cloning systems and create opportunities for significant identity theft, creating deceiving fabricated content and creating elaborate fraud schemes to businesses that rely on the audio of a person for their identity to verify.

Consequently, professionals such as computer scientists, engineers, IT (Information Technology) specialists/ security experts, and machine learning specialists will continue seeking ways to be able to determine whether an audio recording is authentic in real-time or as close to real-time as possible.

A significant body of literature has been published on how voice recognition has changed over time from an analytical and acoustic basis (auditory based) towards more advances techniques such as deep learning and signal processing. The early stages of research into voice recognition relied on handcrafted features, e.g. MFCC, spectral variance, formant frequency and prosody, to describe very small distortions associated with AI generated voice samples. This limited the researcher's ability to assess the accuracy of the features because there was relatively little technology available to generate models and improve upon those distorted features, until the advent of improved technology and methods to generate reliable models that could eliminate the small defects present in AI generated speech.

Over the years, fake voice detection has developed significantly as a result of the rapid development of artificial intelligence and speech synthesis technology. Early research into fake voice detection sought to identify simple acoustic features in machine-generated speech (voice), which were relatively easy to detect because earlier synthesis models had limitations. With the advent of deep learning architectures, however, the quality of synthetic audio has improved dramatically making it significantly more difficult to detect fakes.Throughout the evolution of this area, the shift from traditional signal-processing techniques has also transitioned to advanced machine learning and deep learning techniques. The purpose of these techniques is to extract not only surface-level acoustic characteristics from the speech signals but also deeper patterns that may exist within those signals. This transition has played a large role in improving the accuracy and robustness of fake voice detection systems, especially in real-world cases where it is impossible to avoid noise, and there is a lot of variation.The following table provides an overview of current and primary research pertaining to the project report which relates to the objectives included in this report and will help support the methods and concepts presented within this report.

[1]	Alam (2020)	To improve classification accuracy and robustness against unknown synthetic attacks.	Combined image-based features (Spectrograms) with the highly efficient Random Forest (RF) classifier.	Achieved high accuracy (e.g., 99.93%) in artificial speech detection, validating RF's efficacy.
[2]	Borodin et al. (2024)	To achieve state-of-the-art performance in spoofing detection using graph-based models.	Utilized Graph Attention Networks (GATs) in the AASIST model architecture.	Achieved extremely low minDCF scores (SOTA performance), but possesses the highest computational complexity, rendering it impractical for low-latency deployment.
[3]	Boyd (2023)	To investigate deep learning methods for both binary and multiclass voice spoofing classification.	Comparative analysis of CNN, WaveNet, LSTM, and GRU models using a large dataset.	Confirmed that a robust, unified, and generalized approach applicable in diverse real-world, cross-domain scenarios is still needed.
[4]	Davis & Mermelstein	To establish a feature set for efficient speech processing systems.	Defined the formula and process for calculating the Mel-Frequency Cepstral Coefficients (MFCCs).	Established MFCCs as the canonical features for speech processing, mimicking the human ear's spectral perception.
[5]	de Souza et al. (2025)	To evaluate the impact of feature space reduction on supervised ML models' performance.	Applied dimensionality reduction techniques to feature vectors before training ML models like Random Forest.	Demonstrated significant performance gains for supervised ML models, despite the resulting EER being higher than SOTA Deep Learning models.

[6]	Delgado et al. (2021)	To evaluate the generalizability of Deep Learning models against emerging attacks.	Introduced advanced neural baselines like RawNet2 and LFCC-CNN for the ASVspoof 2021 Challenge.	Confirmed that detection generalization remains an open problem, and demonstrated the high-resource nature of SOTA models.
[7]	Jung et al. (RawNet2)	To develop a dedicated, deep neural network for handling raw waveforms in anti-spoofing.	Introduced the RawNet2 encoder, an advanced deep neural network architecture.	Established a strong baseline for robust speaker verification and spoof detection, but demands significant computational resources.
[8]	Kamel et al. (2025)	To conduct a comprehensive review of DL security solutions, focusing on efficiency and generalization.	Systematic Literature Review (SLR) focusing on ML/DL applications in security.	Confirmed that DL models dominate SOTA but suffer from high computational cost and generalization failures against unseen attacks.
[9]	Khoury et al. (2012)	To model genuine and spoofed speech probability for	Used Gaussian Mixture Models (GMMs) for statistical	GMMs served as early baselines for spoof detection but perform poorly against high-
[10]	Liu et al. (2021)	To eliminate the need for hand-crafted feature extraction by processing audio directly.	Proposed RW-ResNet (Raw Waveform ResNet) to process raw audio waveforms	Showed substantial improvement over traditional LFCC-GMM baselines, but the method is computationally complex.

[11]	Park & Lee (2022)	To review the rapid development and application of Deep Learning techniques for spoofing detection.	Systematic review of DL- based methodologies in biometrics and sensor systems.	Highlighted that while DL provides high accuracy, the computational cost and complexity remain the deployment barrier for real-time systems.
[12]	Patil et al. (2018)	To identify new acoustic features that reliably differentiate synthetic speech from genuine speech.	Derived and extracted features from the phase spectrum of audio signals.	Features derived from the phase spectrum are better at detecting synthetic speech due to phase inconsistencies introduced by TTS/VC algorithms.
[13]	Recent Research (Polynomial Kernel)	To maximize the detection boundary separation in the feature space for artificial speech.	Implemented and tuned an SVM classifier with a polynomial kernel.	Confirmed that using the polynomial kernel in SVM produced a low Equal Error Rate (EER) (below 3%) for artificial speech detection.
[14]	Rao et al. (2024)	To leverage residual learning for highly accurate spoofing detection on standard datasets.	Used a deep ResNet152 architecture to classify spectrogram images on ASVspoof 2019 data.	Achieved high testing accuracy (92.2%), but confirmed the unsuitability of heavy architectures for lightweight CPU deployment.
[15]	CNN-RNN Hybrids	To capture both local spectral and long-range temporal dependencies in speech.	Combined CNNs (for spectral features) with LSTMs/RNNs (for temporal features).	Demonstrated robust detection against disrupted prosodic features, but introduced high architectural complexity and demanding hardware requirements.

[16]	Sonawane & Inamdar (2017)	To compare linear and non-linear classification methods for synthetic speech detection.	Employed Support Vector Machines (SVMs) with MFCCs to classify genuine and spoofed speech.	Non-linear SVM with MFCCs outperformed linear SVM for Synthetic Speech Spoofing Detection.
[17]	Todisco, Delgado & Evans (2017)	To develop anti-spoofing features robust to channel effects and replay attacks.	Proposed and tested Constant Q Cepstral Coefficients (CQCCs) as an alternative to MFCCs.	CQCCs demonstrated enhanced robustness against channel distortion and compression, particularly for Replay Attacks.
[18]	Tsiakoulis & Makris (2023)	To analyze vulnerabilities and attack strategies against Automatic Speaker Verification (ASV) systems.	Review of attack vectors including voice conversion (VC), speech synthesis (SS), and adversarial attacks.	Confirmed that current ASV systems are fundamentally vulnerable as they prioritize identity verification over authenticity/liveness checks.
[19]	Yamagishi et al. (2019)	To establish unified protocols and metrics for spoofing countermeasure development.	Organized the ASVspoof 2019 Challenge, defining new datasets, attacks (TTS, VC, Replay), and the	Unified attack methodologies and included high-fidelity spoofs, necessitating highly accurate, albeit computationally demanding, models.
[20]	Yang et al. (2014)	To apply advanced visual feature extraction techniques to audio signals for classification.	Implemented Convolutional Neural Networks (CNNs) by treating speech spectrograms as 2D images.	CNNs became dominant by extracting intricate spatial-temporal patterns from the spectral domain for detection.

[21]	Zhang & Li (2023)	To review CNN- based methods and frequency-domain feature importance in anti-spoofing.	Literature review focusing on Deep Learning approaches for presentation attack detection (PAD).	Confirmed the critical role of frequency-domain features for detecting AI-generated artifacts and highlighted the challenge of generalization.

The literature indicates that the deep learning models excel at achieving top performance, however they do so at a considerable expense in terms of computational complexity and resource usage. Alternatively, traditional machine learning models when used in conjunction with model-specific feature extraction techniques, provide a more optimal solution with regard to overall performance vs efficiency.

This is what the system is intended to accomplish — balancing the need for accuracy with the need for practicality. The system is built around optimized feature extraction methods (making detection reliable) along with optimized machine learning models, in order to provide reliable detection while being suitable for use in real-time environments.

CHAPTER 3

PROBLEM FORMULATION

3.1 PROBLEM OF AI GENERATED VOICE

Artificial Intelligence (AI) continues to reach new heights, exceeding what both developers and security professionals expected it to achieve. Artificial Intelligence has introduced a new era of higher-quality voice-collision technology (like ElevenLabs, WaveNet, FakeYou, and Resemble AI) that allow people to clone someone's voice using very little original audio content. Although each of these new types of voice-collision technology was originally created with a positive purpose to assist in creating content, provide assistive devices for communications, and improve how "human-like" virtual assistants appear, the recordings produced are so good they now pose a serious security threat.

Artificial Intelligence is also enabling cybercriminals to use cloned voice technologies for illegal purposes. For example, if cybercriminals were able to create a deep fake phone conversation that sounded like a legitimate caller, they may use this technology to steal money from people or trick individuals into giving confidential information needed to bypass authenticating themselves via voice verification on phone systems used by banks and other businesses.

Most of today's voice authentication technologies are based on determining whether or not the sampled voice is the same person as an already enrolled voice sample. This places an immense burden upon the already strained security solutions available today, especially in light of the fact that there are millions of readily available voice duplication tools worldwide. Subsequently, and given the fact that anyone with access to a general-purpose computing environment can create near-perfect copies of any given person's voice; this massively undermines the central philosophy of identity verification-based solutions. Consequently, new questions regarding the operational soundness and security of voice authentication technology have arisen.

One of the primary constraints within the industry today is that contemporary voice authentication systems primarily use the computation of large-scale, state-of-the-art detection methods (such as complex neural networks), which requires considerable amounts of computing power and resources to create and operate a proper model, and are usually only viable on dedicated servers with enough processing capacity to execute all required calculations within a reasonable amount of time. In addition, while rapid development is always possible, specialized applications almost always perform poorly or are untestable, and usually only perform well with only a very limited number of known vocal duplication techniques. Given that there is no one detection method that currently exists or could exist that provides sufficient future-proof security from the current and continually evolving state of the art abilities of advanced vocal cloning technology, the cumulative effect of these limitations is to make the future of voice authentication technology appear to be very bleak.

The aim of the current study is to create a simple, fast, and extremely reliable process for definitively determining whether audio signals come from human voices or artificial intelligence (AI) generated sounds. This differentiation proposal includes developing a software countermeasure that integrates into existing voice recognition systems to provide an additional level of security without requiring organizations to replace their current technology. Currently, with deepfake technologies readily available worldwide, organizations and individuals are facing an increasing and ongoing struggle to verify the true authenticity of audio files, prompting researchers to develop a reliable technological resource to assist in meeting this critical need.

At present, there is no universal, reliable technology available for discerning between human-generated audio signals and those produced by AI systems with certainty. The inherent ability to confirm the acoustic liveness (whether the signal was produced by a living organism) or biological origin (whether the signal originated from a human) of an audio signal represents a very real risk to the secure and accurate verification of audio signals for organizations using voice-enabled authentication technologies. Until methods exist to reliably validate the original source of an audio signal, all forms of AI-enabled voice-authentication systems remain vulnerable to highly sophisticated forms of presentation attacks..

The unique physical characteristics of each person result in different vocal patterns making it difficult to accurately use voice technology for authenticating or verifying users. Voice production involves a multitude of physical and mental factors, as well as lifestyle influences; therefore, many different types of voice sounds and/or patterns exist for each voice.

An effective voice detection system must have both the ability to accommodate differences between voices as well as provide protection against being compromised by people impersonating others with their own voice (an example would be spam or fraud). The challenge to developing a voice detection system is to find the right balance so that the system is sensitive enough to detect fake voices or impersonators and thereby reduce the number of times that a system will generate a false negative identity; while also having enough structure and leniency allowing a user who speaks in normal differences to be able to authenticate their identity with the voice conducted. This will result in fewer instances of a system generating a false positive identity.

3.2 PROBLEM STATEMENT

Recent developments in voice generation technologies such as Eleven Labs and Resemble AI make it possible for anyone to produce an artificial voice that very closely mimics real human voices. These technologies open up a lot of new avenues for creativity. However, they also introduce a lot of possible risks associated with their use - mainly through identity theft and voice impersonation.The growing use of voice as a means of authenticating identity (i.e., verifying an individual's identity) is becoming more common on mobile devices, through financial transactions, and in other instances of using digital services. The problem lies in that current systems used for recognizing someone's voice cannot distinguish between a natural voice (i.e.,

produced by a living being) and a synthetic voice (i.e., an artificially generated voice). This means that you can be exposed to a spoofing attack because you are unable to tell the difference between a real voice and a synthesized voice.

Spammers take advantage of the capabilities of the technology used to synthesize a person's voice, allowing them to impersonate someone else for the purpose of gaining unauthorized access to that person or other unauthorized access or control. The affordability of voice synthesis also increases the risks associated with using voice as a means of identifying people.

This project presents a solution to counteract the issues caused by deep fake technology. Specifically, this includes but is not limited to the creation of fake or AI-generated audio recordings using Machine Learning techniques. As audio recordings are processed, features that provide evidence of the natural characteristics of a human voice will be extracted from the audio data through Mel-Frequency Cepstral Coefficients and spectral patterns. Features will be passed through a classification model (e.g. Support Vector Machines and Random Forests) that has been trained to discern the difference between an authentic human voice and an unauthorized AI-generated audio recording.

The solution provided by this project's proposal is not intended to replace existing systems; but to enhance their effectiveness. The two-tiered process (at both an initial point and during standard authentication) increases security for all digital transactions while reducing the risks associated with identity theft through AI-generated audio.

3.3 OBJECTIVE

The objective of this project is to create a tool that can quickly and accurately determine the acoustic authenticity of speech (liveness verification), as a means of preventing fraudulent uses of the voice. With the rise of advanced voice cloning technologies, it has become increasingly difficult for even trained human listeners to evaluate the inherent authenticity of sound. A key requirement for the system being created is to produce results that meet or exceed current performance standards and can be used in a variety of real-world scenarios. The intended solutions for these requirements include being highly accurate in performance, having a small physical footprint (size and weight) for ease of use, and integrating across various platforms, including mobile applications, smart devices (e.g. smartphones, tablets), virtual assistants, and secure consumer authentication transactions [14].

To create a tool that can reliably determine if vocal production comes from a human or synthetic speaking source, there are three main objectives that must be met.

3.3.1. To Study And Understand How Fake Voices Are Made

The primary objective of this project is to gain an in-depth analytical understanding of current technologies for generating synthetic voices through cloning techniques. The understanding of how synthetic voices are generated will provide important information on the types of artifacts and nuances of sound introduced to audio recordings when these recordings are created using these processes. This knowledge will allow for building a reliable voice detection system. There are numerous approaches to generating synthetic voice using AI today, including:

- Deep learning architectures,

- Neural Speech Synthesis,
- Waveform Reconstruction, and
- Spectrogram Inversion.

For example, companies like Resemble AI, and others, are able to create extremely high quality synthetic voiced copies (i.e. voicetone replicas) as a result of the use of technology that can accurately record most of the human-like characteristics of a voice.

By analysing in detail how AI mimics the inherent qualities of human speech through voice synthesis technology, the comparison to how AI functions will be shown.

3.3.2. To Collect A Proper Dataset Of Real And Fake Voices

It is critical that you have an extensive, high-quality training set to build a highly accurate machine learning model. The third goal of developing a synthetic voice dataset is to include actual audio samples from real people's voices and also synthetic audio produced using AI.

To get actual recordings from real person's voices, we are using two sources: public domains (these are freely available) and also by recording some ourselves. Examples of public domain voice libraries include LibriSpeech, VoxCeleb and Mozilla Common Voice. We will strive to create as much ethnic/gender/tonal diversity as we can, by making arrangements for all elements (age/gender/accent/emotion) of our recordings made by people in the real world, as well as maximizing the number of unique voices in our recordings.

The variety of sources provides a variety of different examples and quality of synthetic voices, as well as the full range of synthetic voices in general. There is an even distribution between samples produced by sources from authentic humans as compared to machine-generated samples. As advances in new voice cloning technologies become available, we will have the largest data set to develop our ML model, which will allow us to create an accurate and reliable ML model to distinguish between samples produced from authentic human voices as compared to synthetic..

3.3.3. To Extract Important Audio Features From The Voice Samples

Analysing unprocessed audio files is not an effective use of Machine Learning Models as these types of files can have a very high amount of complex information. The Project has the overall purpose of converting Raw Audio Files into Features that capture either Human or Synthetic Voice characteristics according to their specific features. The aim of the Project is to convert unprocessed soundwave (Soundwave's) into structured Numerical Values that can be input into a Machine Learning Algorithm to create a differentiation between Real (Human) Speech vs AI Generated Speech.

Some Examples Of Features To Extract Include :

- MFCC (Mel Frequency Cepstral Coefficients): These MFCC features are based on how Human Beings Hear Sounds and (to date) have been used for this purpose in many Speech Recognition Applications. The MFCC feature set captures the characteristics of spectral frequencies present in the Synthetic Voices produced by AI. The System can use MFCC features to distinguish the Natural Spectral Characteristics of the Synthetic Voices produced by AI from those of Human speech.

- The Chroma Features are a measure of the harmonic / pitch characteristics of AI-generated audio. AI-generated audio typically produces very consistent pitch patterns as compared to "normal" human audio; therefore, Chroma Features are a useful way to distinguish between human-created and AI-generated audio.

- Zero-Crossing Rate (ZCR): This is a measure of how often the audio wave crosses the zero-amplitude line. It gives an indication of how the waveform of the audio is developing. Because of the waveform generation method used, the ZCR patterns generated by AI voices are usually significantly different from those of a real voice.

- Spectral Contrast indicates the difference in intensity between the strongest and weakest frequency components of the audio. The audio produced by a real human voice will often display variations and fluctuations whereas synthetic AI voices will usually produce spectral energy that is too consistent.

CHAPTER 4

PROPOSED WORK

4.1 INTRODUCTION OF PHASES OF PROJECT

The systematic method for constructing the Fake Voice Detection System is broken down into 3 different but connected phases; Phase 1 is the Preparation of Data and Feature Engineering, Phase 2 is the Training and Selection of Models, and Phase 3 is the Evaluation and Deployment of System [1]. This methodology is used to achieve a reliable classification of authentic versus AI generated voices in approximately 1/10 of a second [1].

4.1.1 Phase 1: Data Preparation And Feature Engineering

The initial phase establishes the high-quality sound database and extracts acoustic features crucial for the machine learning models.

Building And Processing The Database :

The dataset is developed from a broad collection of human recordings (existing and recently recorded from laboratory conditions) and synthetic voices obtained from benchmark datasets and state-of-the-art AI models to allow for more sophisticated spoofing attacks [2]. The final dataset that was used for training was balanced equally as close as possible by including the same number of real (20) and fake (20) voice samples in the dataset (Figure 1). Thus, it was ensured that the model would not be biased toward either type of voice.

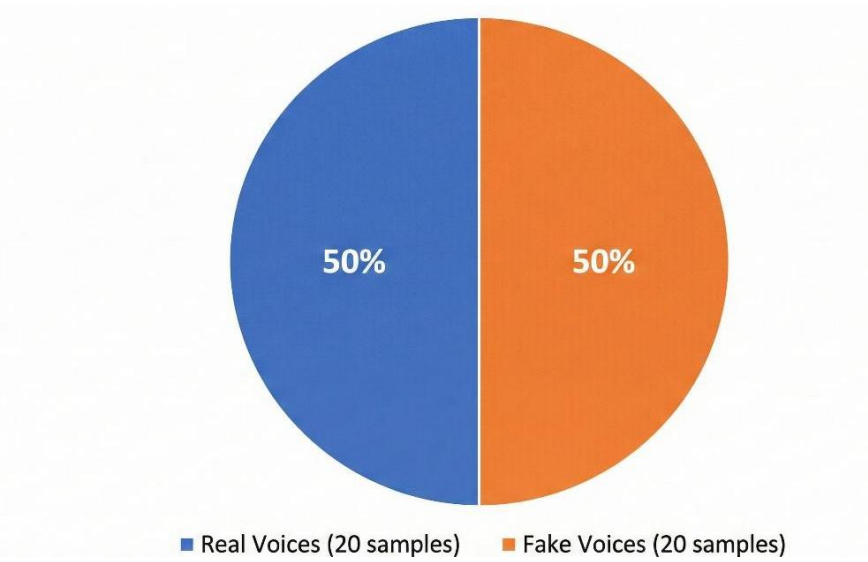

Fig.1 (Dataset Distribution (Pie Chart))

Converting Formats and Sample Rates: The records will be in a single format and sample rate. The purpose of this is to ensure that the models are focusing only on specific vocal characteristics.

Trimming Silence / Equalizing Volume (Amplitude Normalization): Any silence that exists at the beginning and end of the audio records are trimmed off the records. All audio records will be at equal amplitude levels so that there is no bias (due to differences in amplitude) created while training the models using audio records. [3].

Feature Extraction :

MFCC Features
Researchers studying speech have noticed an increase in the popularity of MFCCs (Mel Frequency Cepstral Coefficients). This is because MFCCs closely represent the short-term power spectrum of an audio signal. The original approach used for calculating MFCCs consists of using the discrete Fourier transform to filter the data and then applying a mel-scale to the filtered data. The final step is to take the compressed data and apply a logarithmic function, followed by a discrete cosine transformation; these transformations relate MFCCs to how humans perceive sound [1]. Each speech sample in this study includes 40 MFCC coefficients calculated using a sliding window technique.

Research has shown that MFCCs have a significant impact on identifying differences between human versus artificial speech in terms of both auditory perception and spectral content. When comparing the characteristics of the spectral transitions of real versus deep fake speech, the characteristics of the MG transition from real speech are typically much more gradual than the characteristics of the MG transition from deep fake speech; on the whole, the spectral characteristics of real speech are less likely to be compared with deep fake speech and yet still provide significant differences between the two types of speech.

Additional Acoustic Features
In addition to MFCCs, there are also a variety of other acoustic features that can be extracted from a sound signal, including chroma, zero-crossing rate, spectral centroid, bandwidth, roll-off and root mean square energy. These features describe the voicing characteristics of a voice signal or the distribution of energy within the signal and how that changes because the two signals have different forms of energy or have different distributions from real voice signals to those created with artificial intelligence. Using standard classical machine learning methods, the frame-level features that were extracted will be condensed into a single fixed-length feature vector element by aggregating the features together using statistical measures of the mean and standard deviation. However, in the case of the convolutional neural network (CNN) model, the representation of the original sound will remain intact

4.1.1 Phase 2: Model Training And Selection :

This phase involves implementing, training, and optimizing machine learning models to classify the processed voice samples.

Algorithm Implementation :

Initial development focuses on classical machine learning approaches known for their efficiency with structured numerical features [7]:

Support Vector Machines (SVM): Utilized with an RBF Kernel to find the optimal boundary for classifying data points in the feature space.

Random Forests (RF): Employed as an ensemble classifier, providing robustness and high predictive accuracy by averaging the decisions of multiple individual decision trees [9].

Res2Net-like Convolutional Neural Network (CNN): A deep learning model that learns patterns directly from the MFCC spectrogram built from the audio data [8].

Training And Validation :

The datasets are divided into separate sets for training and testing. The models will learn to correlate the acoustic features of the audio files to the labels associated with them (i.e. authentic/synthetic).

Hyperparameter Tuning is performing adjustments on the hyperparameters that influence the outcome of the model to achieve optimal performance and reduce error rates.

Cross-validation is used extensively to improve confidence in the model and to improve generalisation ability on other previously unseen audio files [10].

The Res2Net-inspired convolutional neural network training completed around a 10-epoch model. Figure 2 presents a visual representation of the training and validation accuracy of the network during each epoch. As shown, the training accuracy and validation accuracy for the network began to increase rapidly until around the fifth epoch, at which point the accuracy began to stabilize at about the eighth epoch. The final validation accuracy achieved approximately 85%.

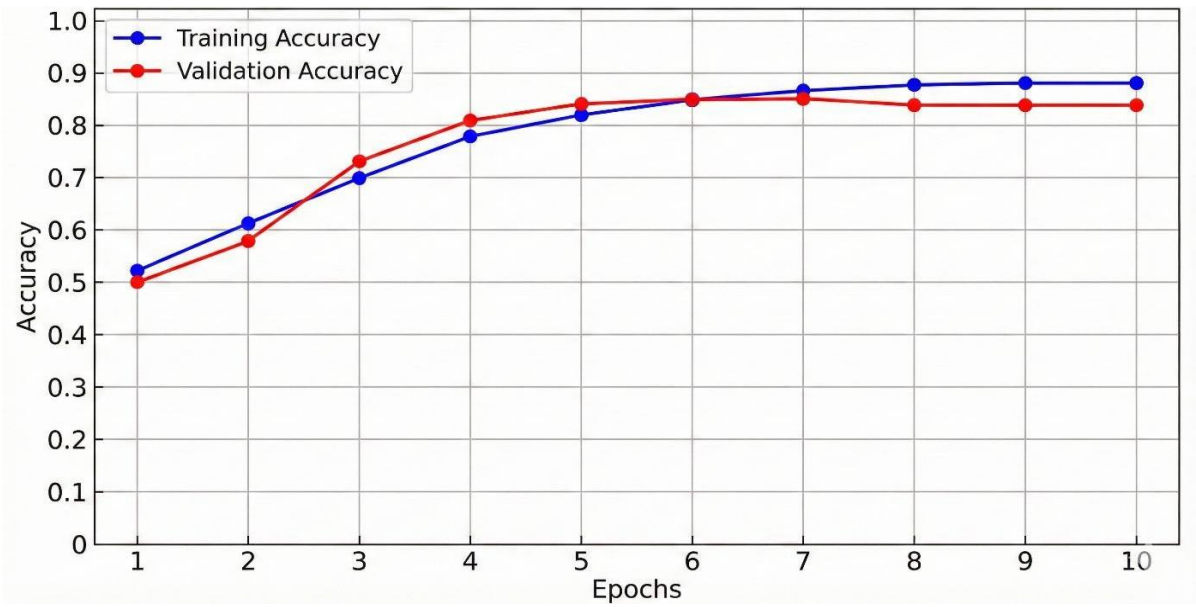

Fig.2 (CNN Training and Validation Accuracy Over Epochs)

4.1.2 Phase 3: Evaluation And System Deployment

During the last stage, we evaluate whether the model is reliable and sufficiently developed for actual use (where it can produce good results quickly). We conducted a thorough evaluation of model performance. The final exam involved an assessment of each model using standard measures of performance (accuracy, precision, recall, F1 score, and confusion matrix), measured both quantitatively (numerically) and qualitatively (how they performed in the environment in which they will be used).

All models then participated in an evaluation under actual use conditions (such as having background noise, a mixture of recording devices, etc.). The Performance Comparison of Model Accuracy (Fig 3) and Summary of Model Performance (Table 1) show the most important conclusions from this research.

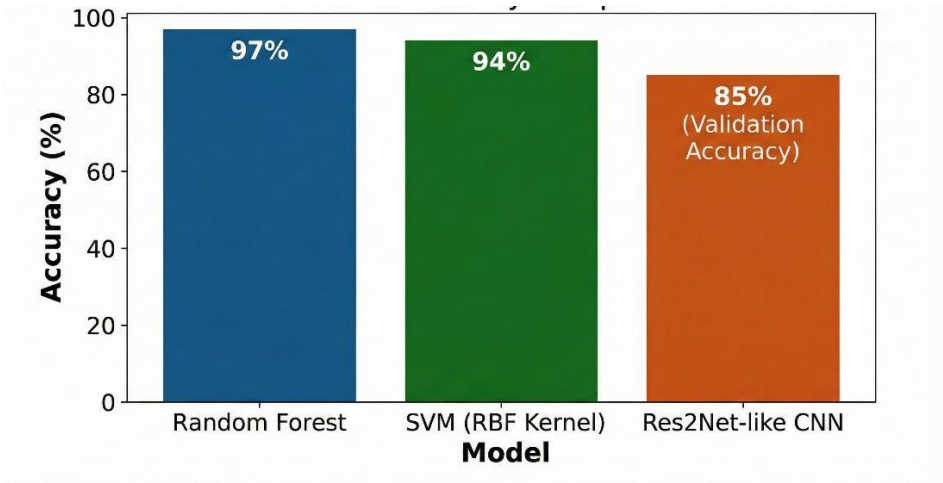

Fig.3 (Model Accuracy Comparison (Bar Chart)

Based on these results, the Random Forest model emerged as the most accurate (97%) and efficient classifier, making it the primary candidate for deployment.

4.2 ALGORITHM :

The following algorithm describes the step-by-step procedure used to train a Convolutional Neural Network (CNN) model for detecting deepfake voice samples. It explains how audio data is processed, transformed into features, and used for model training.

Algorithm: Deepfake Voice Detection using CNN

Input: Real Audio Folder, Fake Audio Folder

Output: Trained CNN Model

```
BEGIN
real_data ← Load_Audio(Real_Audio_Folder, label = 0)
fake_data ← Load_Audio(Fake_Audio_Folder, label = 1)

dataset ← Combine(real_data, fake_data)

FOR each audio_file IN dataset DO
   mfcc ← Extract_MFCC(audio_file, 40)
   mfcc ← Pad_To_Max_Length(mfcc)
   Store(mfcc)
END FOR

X, y ← Prepare_Dataset()
X ← Add_Channel_Dimension(X)

X_train, X_test, y_train, y_test ← Train_Test_Split(X, y)

model ← Build_CNN_Model()
Train(model, X_train, y_train)

Save(model)
RETURN model
END
```

4.2.1 Explanation of Algorithm Steps :

Step 1: Data Loading

The process of creating an algorithm starts with gathering instances of audio sound files. There are usually two different types of audio data: real voice audio data and synthetic voice audio data. The real voice audio files are those that come from real people, while synthetic means they originated from some type of computer generated source. Each instance of an audio file has a corresponding label. For example: real would have a label value of 0 and synthetic would have a label value of 1. Labeling will benefit the algorithm by allowing the model to use these label values while training to distinguish between the two classes of audio data.

Step 2: Dataset Formation

Once the two classes have been distinguished, the respective voices from both classes are then merged into another single "Training" Dataset. The purpose of this singularly created dataset is for the model, through learning, to receive both of the audio datasets at the same time. By separating out real and synthetic voice audio datasets during the initial stages of creating an algorithm, there will be no uncertainty regarding which dataset belongs to appropriate category. It is very important to have the labels correct as the model relies on the accuracy of the labels to learn.

Otherwise there could be errors when it comes time for prediction during the test phase. Moreover, if these errors occur, the model's predictions cannot be trusted, and thus the whole algorithm fails. For these reasons, it is critical to maintain accurate and consistent data between both classes in order to provide the model with a trustworthy training data set.

Step 3: Features Extraction

For each audio file within the data set, we will compute Mel-Frequency Cepstral Coefficients (MFCCs). MFCCs have been widely used in previous research pertaining to audio analysis, since they adequately describe how humans speak. Due to varying audio file lengths between files, all files have been padded to create the same sized inputs to the model.

The process of Feature Extraction transforms raw audio inputs into numerical representations, providing one of the steps toward preparing audio signals for the model. MFCCs represent a compact and weighted representation of the characteristics of audio signals rather than using the complexity of the raw waveform.The transformation of these audio signals into MFCCs enables the model to concentrate on patterns necessary to differentiate between natural and synthetic speech.

Step 4: Data Preparation

Once the features have been extracted, the values are stored as input features (X) and corresponding labels (y). Additionally, one more channel dimension will be added to the input feature data to be compatible with a CNN input, as CNN models expect multi-dimensional input.

Uniform dimensions throughout the training data are essential for smooth training of the model. All audio files may contain various lengths of time, therefore the length of each file will also vary; padding introduces uniformity of the input sizes, and allows the training dataset to be processed uniformly. As a result of uniformity in the dimension of the training dataset there are fewer complications with computation and compatibility with configuration of convolutional neural network algorithms.

Step 5: Data Splitting

The dataset has to be split into two subsets: one for training (the training set) and one for testing (the testing set). The training set is to be used to fit the model's parameters, while the testing set is to be used for evaluating the model's performance on previously unseen data points. The appropriate separation of a dataset into training and test sets is essential to assess the generalisation ability of the generated model; an appropriately trained predictive model should produce acceptable performance levels not only on the training sample but also degree of accuracy on new data.

The very first of these steps to ensure overfitting does not occur and thus confidence that the predictive model would be reliable in practical applications.

Step 6: Model Building

A CNN model will be built. CNNs are identified as very successful in completing types of tasks requiring a recognition of pattern among a significant dataset (e.g. CNNs and identification of categories from one type of specimen to another); similarly when quantifying an audio file through a visual representation that resembles a spectrogram, CNNs can correlate numerous levels of complexity within the relationships of audio data.

Step 7: Model Training

The model can now be trained with the prepared training data; at this stage, the CNN will be able to distinguish between real and fake audio samples using identified patterns found within each dataset of audio; therefore, during the training process of the model, the model will be trained multiple times by making internal adjustments to improve upon and reduce the number of prediction errors.

This process of training enables the CNN to detect small differences between each type of audio signal (real versus fake) by identifying differences in frequency patterns, tonal patterns, and temporal patterns.

Step 8: Model Saving

The model has been trained and is ready for use. The model can now be used to predict future cases without the need to retrain the model each time; thus saving time in being able to predict without having to go through the lengthy process of retraining the model.

The importance of saving the trained model is to provide the system with the ability to reuse the knowledge learned without having to go through the training of the model a second time.This saves significant time in terms of computing and enables the rapid deployment of the system for use in real-time applications.

Step 9: Output Generation

At this point, we have returned our trained model, which can now be used to classify new audio samples as either real (live recording) or fake (deepfake).

Overall, this algorithm provides a complete pipeline for deepfake Voice Recognition by using a combination of different types of data (audio recordings) to produce an accurate final output model, with each part of the process contributing towards achieving a successful result. The result of combining feature extraction methods with deep Learning techniques form the basis of the ability for the system to identify complex relationships between actual human voices and synthetic generated audio, therefore making this approach suitable for many real-world applications that require ensuring security and validity/credibility of voice recordings.

CHAPTER 5

SYSTEM DESIGN

The system design phase focuses on defining the overall structure, components, and operational flow of the proposed solution. It provides a clear understanding of how different elements of the system interact with each other to achieve the desired functionality. A well-defined system design ensures that the implementation process remains organized, efficient, and scalable.

5.1 FUNCTIONAL SPECIFICATION OF SYSTEM

5.1.1 Hardware Requirements

This project does not need any special or costly hardware, but having a good computer makes the work faster and smoother, especially when training machine learning models and processing audio files. The following hardware components were used during the development of this project:

Processor (CPU) :

A multi-core CPU has to be chosen, either an Intel i5/i7 processor or an AMD Ryzen 5/7 CPU. A good CPU is necessary to perform many functions, such as reading audio files, cleaning audio files, & Feature extraction etc.

Graphics Card (GPU) :

A GPU is very beneficial for training deep learning models, like CNN's. Deep learning models take much longer to train if you are only using a CPU, but using a GPU will speed up the process significantly. In addition, TensorFlow & Pytorch have better performance on GPUs that support CUDA, such as Nvidia GTX & RTX series. A GPU is not required, but they help speed up the training & create a better experience during development.

Memory (RAM) :

A minimum of 8GB of RAM must be available to run applications. RAM will allow the data to be retained temporarily while the program is operating. In order to work with an audio sample dataset and machine learning models simultaneously, a high-enough RAM must be available to prevent lagging or slowdowns. A system with more RAM will allow for larger datasets without having to deal with the lag time.

Storage (SSD) :

A dataset, audio files, spectrogram images, trained model, and log files were stored on either a 500GB or 1TB Solid State Drive (SSD). An SSD has a greater speed when compared to a traditional hard disk drive (HDD).

Microphone / Audio Recording Device :

This project requires the collection of real human speech; therefore, a high-quality microphone was selected to obtain the best recordings. All audio recordings were made under very controlled

conditions to minimize background sounds and to standardize the data.

5.1.2 Software Requirements

Additional to the hardware requirements for the system, there were also many software tools, libraries, and frameworks that were used to create and evaluate the finished product; The programming language that was utilized for the entire project was Python.

Programming Language – Python :

Python is chosen as the programming language due to its extensive set of powerful Libraries for Machine Learning; Audio Processing; Web Development; and overall; has easy to read/learn Syntax, resulting in a less time consuming and easier Development Process.

Libraries and Tools Used :

Machine Learning and Deep Learning Libraries.

- Scikit-learn – The Scikit-Learn Library is utilized while utilizing different algorithms such as Random Forest; SVM etc. for different types of Machine Learning Models.

The above-listed libraries and tools provided many advantages regarding training and evaluating different models in a timely manner.

Audio Processing Libraries

- Librosa – Librosa is a python library designed for audio and music processing and analysis.

Data Handling and Visualization Tools

- NumPy – Used for numerical operations and handling arrays.
- Matplotlib – Used to create graphs like accuracy curves, confusion matrices and similar plots.
- OpenCV – Used for processing spectrogram images before sending them to the CNN model.

GUI Tools

A user-friendly System was created by developing a gui.

- PYQT5 – PyQt5 is a Python binding of the cross-platform C++ framework Qt v5, developed by Riverbank Computing Limited. It enables Python developers to create graphical user interface (GUI) applications.

This allowed users to upload immediately see whether the voice was real or fake.

Version Control Tools

- Git – Maintains a record of changes to the source code and maintains various versions.
- GitHub or GitLab – Enables backup and secure storage of the project's source code via online

repositories.

These tools helped maintain an organised workflow.

Development Environments (IDEs)

- VS Code – An IDE for writing and debugging the main Python code for the project.
- PyCharm – A dedicated IDE for the creation and testing of the Machine Learning code for the project.
- Jupyter Notebook – Provides a convenient environment for experimentation, prototyping, and visualising results incrementally.

5.2 STRUCTURAL AND DYNAMIC MODELLING OF SYSTEM

5.2.1 Activity Flow

The activity diagram of the fake voice detection system described below outlines the workflow of converting audio from user input through to the final detection result. The workflow is described in the following way:

- An Overview of the Process from audio input through to the detection result is shown here, including audio validity checking, audio pre-processing (trimming/padding), MFCC feature extraction, and model running to make detection predictions.

- The diagram depicts the basic logic of how a fake voice detection system, through the pre-processing stage of audio input, is able to transform raw audio into MFCC features that are meaningful to the system and provide the basis or foundation for classification of the audio through ML models.

- The system will load the input audio file into the fake voice detection system. The output from this stage (pre-processing) is standardised audio processed using trimming (eliminating parts of the audio from either end of the audio before processing) and/or padding (adding blank space at either end of the audio to bring clipped samples up to a standardised audio length).

- Standardisation of audio provides a consistent and uniform audio sample on which to base the detection analysis. Without a consistent audio sample, the analysis of the audio sample would result in inconsistent audio output during the ML detection model running stages.

- After standardisation, the audio file is then processed for feature extraction using the Mel Frequency Cepstral Coefficients (MFCCs). The MFCCs derived from the audio signal contains all the information necessary to define the characteristics of human speech, including frequency and tonal characteristics. The audio data is now represented in a structured numerical format and is suitable for use by ML models for the classification of fake voices.

- The overall workflow depicted in the activity diagram illustrates the audio processing flow from raw audio to cleaning, normalisation and transformation prior to its use as a detection data source for fake voice detection.

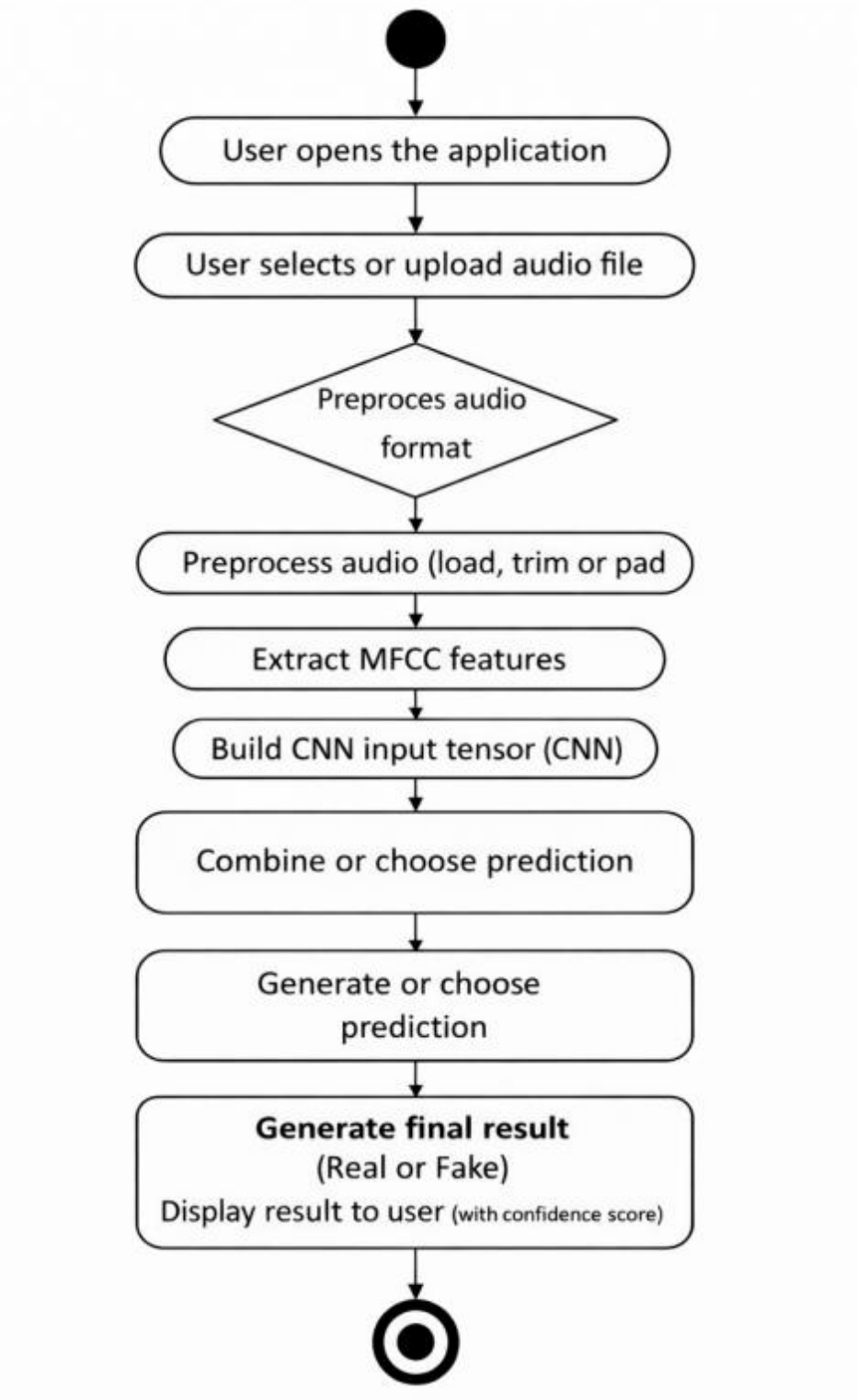

Fig.4 (Fake Voice Detection System - Detection Workflow (Activity Diagram))

5.2.2 Sequence of Operations

The Fake Voice Detection System - Detect Workflow sequence diagram (Fig. 5) details the interaction between system components when a user clicks 'Detect'.

The sequence diagram illustrates how different components of the system interact when a user initiates the fake voice detection process. The process begins with the VoiceAuthGUI, where the user selects and submits an audio file for analysis. This request is then forwarded to the PredictionService, which acts as the central controller of the system. The PredictionService sends the audio file to the AudioPreprocessor, where the audio is cleaned, normalized, and adjusted (such as trimming or padding) to ensure consistency. After preprocessing, the audio is passed to the FeatureExtractor, which extracts important audio features such as MFCCs that represent the characteristics of the sound. These extracted features are then forwarded to both the DeepModel and the ClassicalModel, where predictions are made based on learned patterns. Once the models generate their outputs, the PredictionService collects the results, determines the final classification (real or fake), and sends the result back to the VoiceAuthGUI. Finally, the GUI displays the prediction outcome to the user in an understandable format. This sequence ensures a smooth and structured flow of data across all components, enabling accurate and efficient fake voice detection.

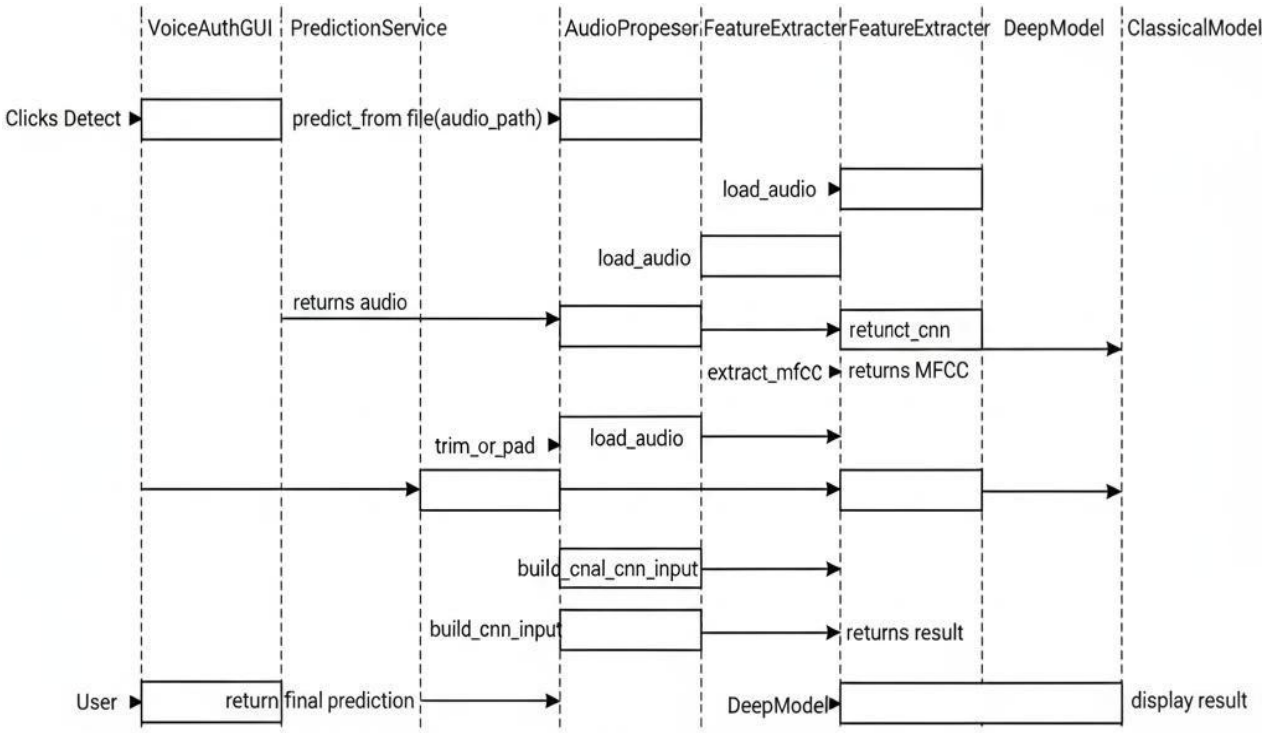

Fig.5 (Fake Voice Detection System - Detect Workflow (Sequence Diagram))

5.2.3 System Architecture

The Fake Voice Detection System class diagram (Fig. 6) shows the relationships and dependencies between key software components: VoiceAuthGUI relies on PredictionService.

The class diagram represents the structural design of the fake voice detection system by illustrating the relationships between its core components. The system is centered around the PredictionService, which acts as the main processing unit responsible for coordinating different modules. The VoiceAuthGUI interacts directly with the user, allowing them to select an audio file and trigger the detection process. Once initiated, the request is passed to the PredictionService, which utilizes the AudioProcessor to load and preprocess the audio by trimming or padding it to a consistent format. The processed audio is then forwarded to the FeatureExtractor, which extracts MFCC features and prepares inputs for both classical and deep learning models. The system includes two types of models: the Deep Model, which uses a CNN for advanced pattern recognition, and the ClassicalModel, which relies on traditional machine learning techniques along with a scaler for feature normalization. Both models generate predictions, which are then managed by the PredictionService and returned to the GUI for display. The diagram clearly shows dependency and interaction between components, ensuring a modular and well-organized system architecture.

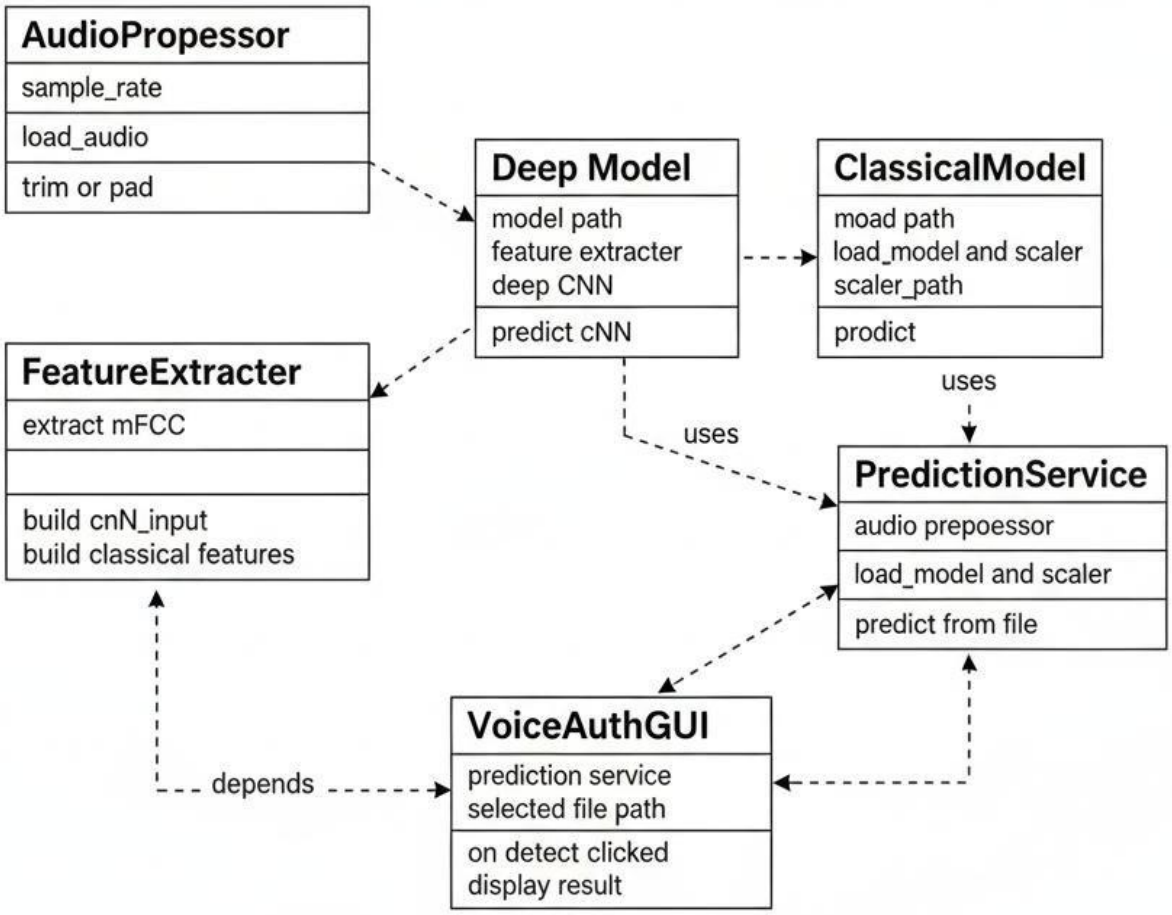

Fig.6 (Fake Voice Detection System (Class Diagram))

5.2.4 Use Case

The system supports core functionalities for two primary actors, as shown in the Use Case diagram (Fig. 7): User (Upload Audio File, Run Fake Voice Detection) and Admin (View Detection Result, Manage Dataset).

The use case diagram represents the interaction between different users and the fake voice detection system, highlighting the core functionalities provided by the system. There are two primary actors: the User and the Admin. The User interacts with the system by first uploading an audio file and then initiating the fake voice detection process. Once the analysis is complete, the system processes the input and generates a result. The Admin, on the other hand, has additional privileges, including the ability to view detection results and manage the dataset used for training and testing the models. This includes tasks such as updating, adding, or maintaining audio data. The diagram clearly defines the roles and responsibilities of each actor, ensuring that system functionalities are properly organized and accessible based on user permissions.

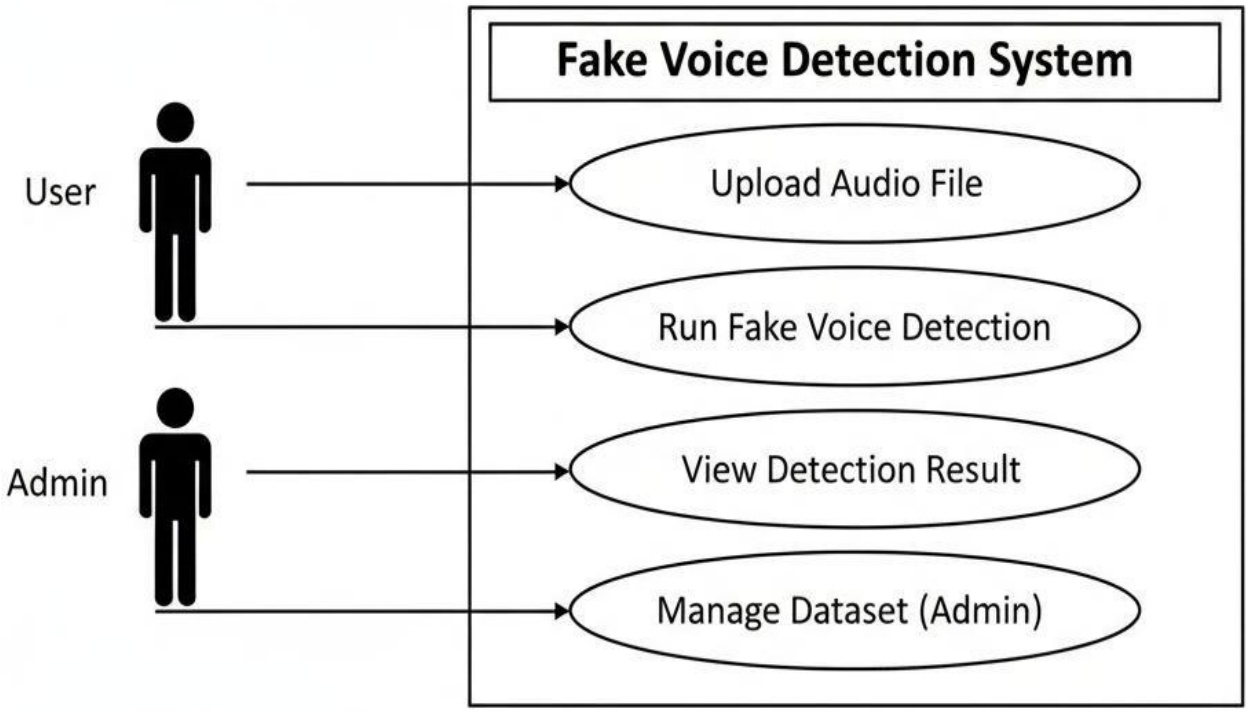

Fig.7 (Use Case Diagram)

CHAPTER 6

IMPLEMENTATION

The implementation phase converts the theoretical concepts and proposed methods into a functional system. The implementation phase is a key phase to verify the proposed feasibility of the approach and to ensure that the system meets the required performance under real-world conditions. Therefore, the balance of performance, efficiency and usability has been carefully maintained to ensure that the system can be used in a real-world environment without the need for significant computational resources.

The implementation is also modular, making it flexible and scalable. All of the elements of the system such as feature extraction, model training and prediction operate independently of each other yet contribute to the overall workflow of the system. This modularity also contributes to the ease of debugging and testing, while allowing for future modifications to be made without affecting the overall architecture of the existing system.

VoiceAuth combines regular machine learning techniques with some experimental approaches to deep learning, as well as an initial desktop GUI (graphical user interface). This entire collection of components establishes a complete system designed to assist users when identifying AI-produced audio (commonly referred to as 'deepfake' audio). Throughout this chapter, we will describe the overall structure of the VoiceAuth system, how the various components interact (how data flows through the various components), and provide a detailed, step-wise representation of the algorithm that is executed during both training and production use of the VoiceAuth system. The narrative will illustrate how each of the parts of the VoiceAuth system fits together (from a functional perspective) and identify the specific functionality performed by each module; i.e., feature extraction and classification is performed by the classifier.py module, GUI-related user interactions are processed through the gui.py module, and an optional deep learning module is implemented using the back.py module. The complete description of the algorithm provides both clarity and ease of replication and provides an excellent understanding of how the VoiceAuth system operates in day-to-day use.

6.1 IMPLEMENTATION AND EXPERIMENTAL SETUP

6.1.1 Project Layout

The organization of the project is dependent upon the directory structure, which helps separate different functions of the system. A well-defined project layout allows the codebase to be kept clean and easy to navigate for future development. Each folder and file has been given a purpose so developers can easily locate where a specific function has been coded.

Additionally, the well-structured format allows for growth because you can add new modules or enhancements without interrupting other parts of your workflow. And since several people can simultaneously contribute to different pieces of the system, this increases productivity as well.

In addition, the way components were separated results in some components not being impacted by changes to other components. As seen in Fig. 8, this allows independent testing or debugging of isolated components because they operate independently. The enforced separation between components also facilitates ease of adding advanced models and functionality; therefore, it will not require re-engineering the entire system if new models or functions need to be integrated later

on.

```bash
VoiceAuth/
├─ data/
│  ├─ AUDIO/
│  │  ├─ REAL/
│  │  └─ FAKE/
├─ models/
│  ├─ rf_model_v1.joblib
│  └─ scaler_v1.joblib
├─ gui.py                # PyQt5 desktop GUI
├─ classifier.py         # Lightweight ML pipeline (feature extraction + RF)
├─ back.py               # Experimental deep-learning pipeline (Res2Net-ish)
├─ utils.py              # shared helpers (io, logging, audio utils)
├─ requirements.txt
└─ README.md
```

Fig. 8 (Project Directory Structure)

In order to understand the details about each library being used for the design implementation of the overall audio experiment system, it is first necessary to appreciate how the system combines three very different fields of expertise: audio signal processing, machine learning, and graphical user interface design; each of these areas is represented in the libraries selected for implementation. Furthermore, the type of libraries chosen for implementing this system was based on their utility and ability to accomplish the desired functionality of the overall implementation.

6.1.2 Key Libraries

- librosa — audio loading & feature extraction (MFCCs, chroma, mel).
- numpy, scipy — numeric & signal utilities.
- scikit-learn — RandomForestClassifier, StandardScaler, model persistence (joblib).
- joblib — save/load model and scaler.
- PyQt5 — GUI (buttons, file dialog, progress bar, message boxes).
- tensorflow / keras — optional deep model in back.py.
- soundfile / pydub — optional audio I/O / normalization.

By using the libraries mentioned above, the end-to-end processing that takes audio as input and outputs predictions is done effectively through the use of specific libraries with different functionalities. For example, librosa was used to extract features from the audio, while scikit-learn was used to create efficient machine learning models; additionally, an interactive interface for users was created using PyQt5.

By following a modular organization structure that consists of functional components within the pipeline, the overall implementation can provide both clarity and modularity. Each of the various modules is responsible for a particular function or task, which keeps the implementation organized and manageable.

6.1.3 Important Modules & Functions

classifier.py (core, lightweight pipeline) :

The classifier module is the core component of the system, as it manages the complete workflow of data processing and model execution. It ensures that audio data is properly transformed into feature representations before being used for training or prediction. The use of separate functions for feature extraction, dataset collection, training, and analysis improves code readability and reusability. This structured approach also simplifies debugging and allows individual components to be tested independently without affecting the entire system.

• extract_features(path: str, duration: float = 10.0) -> Optional[np.ndarray]

Loads audio (librosa.load), limits duration, computes:

1. MFCCs, Chroma, Mel spectrogram
2. For each feature matrix compute statistics per coefficient: mean, std, max, min
3. Concatenate into 1D vector and return
4. Returns None on error (IO or corrupt file)

• collect_dataset(real_dir: str, fake_dir: str) -> (X: np.ndarray, y: np.ndarray)

Walks directories, extracts features for each file, builds labels (0 = REAL, 1 = FAKE).

Skips files where extract_features returned None.

• train_model(real_dir: str, fake_dir: str, out_model_path: str, out_scaler_path: str, n_estimators: int = 200)

1. Calls collect_dataset.
2. StandardScaler().fit_transform(X) → X_scaled.
3. Trains RandomForestClassifier(n_estimators = n_estimators, n_jobs = -1) on X_scaled, y.
4. Persists classifier and scaler using joblib.dump(...).
5. Returns training metrics (accuracy on OOB or via cross-validation) and saved paths.

• analyze_audio(file_path: str, model_path: str, scaler_path: str) -> dict

1. Loads model & scaler via joblib.load.
2. Extracts features, scales, uses model.predict_proba to compute p_fake, p_real.
3. Returns JSON-like dict:

   { "is_fake": bool, "fake_probability": float, "real_probability": float }

4. Raises exceptions for missing model/scaler or bad audio.

gui.py (PyQt5 front-end) :

The graphical user interface acts as a bridge between the user and the system. It simplifies complex operations such as model training and audio analysis into easy-to-use actions. By providing visual feedback and clear interaction steps, the GUI improves accessibility for users who may not have technical expertise.

- Buttons: Train New Model, Select Audio File, Analyze Audio.
- Workflow:

1. Train New Model → disables button, shows progress, calls classifier.train_model(...) and updates UI on success/failure.
2. Select Audio File → opens QFileDialog; selected path stored.
3. Analyze Audio → shows progress bar, calls classifier.analyze_audio(selected_file), reads result and updates labels/colors/plots.

- Error handling: wrap all long operations in try/except and present QMessageBox for user-friendly errors.
- Note: current implementation blocks the GUI during training/inference; recommended to move heavy work to QThread or concurrent.futures.ProcessPoolExecutor.

Model Versioning & Persistence :

Properly storing the trained model with associated configuration data, including metadata, is critical to enabling tracking of how the model performs through time. By saving the model and its configurations with identifiable names, it is easy to compare training runs and reproduce an experiment when needed.

Examples:

• Save the model and associated scaling with all models listed with their training epoch version number (e.g., rf_model_v1.joblib; scaler_v1.joblib).

• Save all training metadata (train_log.json) associated with the training run which includes the dataset path, timestamp of the training run, feature configuration used in the training run, hyperparameters, and validation metrics.

Error Handling, Reliability, And Tooling Notes :

Robust error handling provides a guarantee that your system will continue to function as designed and service users reliably even with unexpected inputs or failures. By logging potential error scenarios and safely bypassing error causing files, your system maintains the integrity of the processing and eliminates unnecessary interruptions due to potential errors.

Examples

• For every call to extract_features when there is a problem driving the data to the system; it will return None and log file that caused the failure; the system will store the call to extract_features in skipped_files.log when the incoming file has failed.

• When back.py calls extract_features to retrieve the data, intermediate web application generated files are deleted from the defined temporary location by using shutil.rmtree(tempdir, ignore_errors=True).

• Recommended unit testing for extract_features, train_model, analyze_audio.

6.2 DATASET DESCRIPTION :

The dataset for the current study was created from the balanced feature datasets from the FILE - DATASET-balanced.csv file (Bird, 2020). This dataset is the result of recordings taken from the DEEP-VOICE corpus, which was developed by Bird and Lotfi for the purpose of developing AI technologies for the production of synthetic speech, as well as the development of AI technology for the identification of synthetic speech produced by humans in real-time.

The dataset used for this study provides a structureally balanced representation of both authentic and synthetic audio samples. An important factor to developing an effective machine learning model is to ensure that it has a proper structure and balanced dataset that will allow it to learn how to distinguish between authentic and artificially generated voices without introducing any bias into the learning process. In addition, the use of a well-defined dataset is critical to increasing the reliability and consistency of the results produced during the research study.

6.2.1 Dataset Size And Structure

The dataset contains:

• Total Samples: 11,778

• Total Features per Sample: 26 acoustic features

• Label Column: 1 (REAL / FAKE)

• Total Columns: 27

Each row represents a fixed-length feature vector.The fixed-length representation of each sample simplifies the training process, as machine learning models require uniform input dimensions. This structure eliminates the need for complex preprocessing steps during training and allows for efficient handling of large datasets extracted from a 1-second audio segment, as described in the original DEEP-VOICE study [Bird & Lotfi, 2023].

6.2.2 Class Distribution

The class distribution presented in Table 1 shows that the dataset is evenly balanced between real and fake audio samples. This balance is crucial in classification problems, as it prevents the model from becoming biased toward one class. As a result, the evaluation metrics obtained from the models provide a more accurate and fair representation of their performance.

Class	No. of Samples
Fake	5889
Real	5889
Total	11778

Table 2(Dataset Class Distribution)

6.3 PSEUDOCODE

The following pseudocode outlines the step-by-step procedure used in the deep learning-based approach for fake voice detection. It provides a clear and simplified representation of the workflow, making it easier to understand the sequence of operations involved in the system.

ALGORITHM: Deepfake Voice Detection using CNN

INPUT: Real Audio Folder, Fake Audio Folder

OUTPUT: Trained CNN Model

```
BEGIN
real_data ← Load_Audio(Real_Audio_Folder, label = 0)
fake_data ← Load_Audio(Fake_Audio_Folder, label = 1)

dataset ← Combine(real_data, fake_data)

FOR each audio_file IN dataset DO
   mfcc ← Extract_MFCC(audio_file, 40)
   mfcc ← Pad_To_Max_Length(mfcc)
Store(mfcc)
END FOR

X, y ← Prepare_Dataset()
X ← Add_Channel_Dimension(X)
X_train, X_test, y_train, y_test ← Train_Test_Split(X, y)
```

```
model ← Build_CNN_Model()
Train(model, X_train, y_train)
Save(model)
RETURN model
END
```

This pseudocode will describe the overall process of loading audio data into the model, training the model using that audio data, and then predicting with the trained model on input audio. Each step in pseudocode corresponds to an action taken throughout the implementation; e.g., feature extraction, preparing data, and training a model.

Using pseudocode also assists readers in linking their theoretical understanding of these concepts with how those theoretical concepts are instantiated into an actual implementation of a detection system by using pseudocode to help readers understand the logical flow of the system and not be dependent on any specific programming language. Therefore, it is possible to achieve both clarity and reproducibility through the use of pseudocode.The way in which the system is implemented plays a crucial role in producing the performance results that follow. The method(s) chosen for feature extraction, architecture of the model, and approaches to processing the data will significantly impact how accurate and efficient the detection system will perform.

The results provided through the analysis within this study support understanding how effective different machine learning algorithms were to determine if a voice was being faked. High accuracy is only one measure of how well a model performs, other elements also play a critical role in an eventual model deployment; such as how much it costs to run the model, how well the model will scale if additional operational capacity is required, and how easy it will be to deploy the model into an operational environment.

One of the key observations from this study is the trade-off between accuracy and computational complexity. Deep learning models, such as CNNs, are capable of capturing highly complex patterns in audio data, making them effective for detecting advanced synthetic voices. However, their dependency on large datasets and powerful hardware limits their usability in resource-constrained environments.

This study reveals a trade-off between accuracy and the amount of computation needed to derive that accuracy. Deep learning algorithms like CNN's can detect advanced synthetic voices as they are able to characterize very complex patterns in audio data; however their ability to be used in constrained resource environments is limited due to their dependence on large amounts of training data and expensive hardware resources.

CHAPTER 7

RESULT ANALYSIS

Evaluating the effectiveness of counterfeit voice detection systems has been more critical over time with the advent of technology used to create synthcetic voices. As Artificial Intelligence (AI) generates audio more realistically than ever before, basic assessment techniques aren't going to cut it anymore. A thorough analysis of how well detection models work is essential to ensure they function as expected in situations where there are vast differences in audio quality, voice characteristics, and ambient noise levels.

This evaluation aimed to quantify the performance of different machine learning models - more specifically, Random Forest (RF), Support Vector Machine (SVM), and Convolutional Neural Networks (CNN) - when used to distinguish real human voice recordings from AI generated / synthetic voice recordings. Identifying fake voice recordings accurately is crucial to counteracting any potential spoof attacks that could potentially allow an attacker access to an organization's access control or security systems or other forms of economic fraud.

7.1 MODELS USED IN RELATED WORK

7.1.1 Experimental Setup and Feature Efficacy

To provide a fair evaluation of the various methods applied, all model methods were supplied with feature vector representations created from the feature extraction performed during the Pre-processing phase (discussed within Section 8). For this evaluation, the utilized acoustic features included Mel-frequency cepstral coefficients (MFCCs), chromaticity, and spectral contrast. These features were chosen because they provide a rich representation of audio information and allow for the detection of subtle tonal, pitch, and frequency characteristics of natural speech which can often be imperfectly reproduced by synthesised audio. With the inclusion of these wide-ranging feature sets, the system had sufficient input to identify minute differences between true and artificially produced voices.

Finally, for each model to be trained and tested under identical conditions, it was essential that they were all built consistently in terms of data handling, feature extraction, pre-processing and model training. This provided all models with the same amount of data processed in the same way, to ensure that biases occurred as a result of some models being trained with more or different quality data (due to variations in pre-processing).

In addition, the specific feature sets selected for use will have an important impact on the ability of the model to perform successfully. Quality features provide a solid foundation for classification, which can allow even simple models to perform effectively. Thus, feature extraction is a critical component of the overall system design.

7.1.2 Comparative Model Performance Analysis

Qualitative assessments of how models behave also offer a greater understanding of how they perform than their numerical metrics alone, such as examining how they react to audio samples that exhibit characteristics on the borderline or ambiguous side of the classification spectrum helps build an understanding of the robustness of each model. While it is important to know that the models perform consistently, regardless of the categories of the input data, it is equally important to understand whether the model performing well is representative of performance in real-world use, as opposed to in a predictable environment through controlled testing.

The overall performance of the models was compared using standard classification methods across a large number of tested samples to allow for a thorough comparison of their strengths and weaknesses for real-time use.

7.1.3 Random Forest (RF) Classifier

The Random Forest (RF) Classifier produced the best overall average performance among all tested models with an overall accuracy of around 98% and a consistently high range of accuracy from 96% - 98%. The RF model's inherently good performance is due to the RF model's nature of taking an ensemble of predictions from multiple decision trees and using an overall average of the predictions to arrive at the overall prediction, which balances predictive accuracy to computational efficiency.

A major advantage of the Random Forest model is that it is highly resistant to overfitting. As multiple decision trees are combined to create the final prediction for each test instance, the RF model is much less likely to overfit to a particular pattern in training data than are individual decision trees. This means that the Random Forest model is able to generalize to newer, previously unseen audio samples with a high degree of accuracy and should be highly dependable when applied in real-world situations.

The Random Forest model also provides additional value when it comes to its interpretation of and analysis regarding how it determined its classifications (notable in its ease of analysis of features used to classify). Because it is a model that relies on classification rule learning instead of the complex deep learning methods, as such, understanding features that contributed the most to the final classification decision made by the model has been beneficial when it comes to security related applications where understanding how and why the prediction was made, is critical to security objectives.

7.1.4 Support Vector Machine (SVM)

The Support Vector Machine (SVM) model assigns a point on a hyperplane so as to separate the classes of data being analyzed into two discrete classes, as shown in the image. Therefore, SVM models provide a next-best result for the identification (classification) of genuine and synthetic voices, with an average accuracy level of approximately 94%. The SVM's accuracy level is only slightly lower than that of the Random Forest's accuracy level, however, it still provides a reasonably high level of confidence in classification results that can be used as a basis for the

identification of voice types. The hyperplane generated for classification of voice types is based upon the defined high-dimensional space of features (real-world factors) upon the classification problem; however, the computational resources required to generate such a hyperplane for the SVM may be greater than those required to generate a Random Forest model.

7.1.5 Convolutional Neural Network (CNN)

The Convolutional Neural Network (CNN) demonstrated great promise in its ability to discriminate between even the slightest variations in the audio signal. The CNN processes the spectrogram of the audio signal (treating it as an image) to automatically extract complex visual patterns that are difficult to discern from numerical feature representations alone.

Because of this ability of the Conv-Network, it was found to be one of the best solution for detecting complex fake audio created by new AI technologies for recreating human emotion, sound of natural pauses in human speech and complexity of speech characteristics. The results indicated that for many cases, the Conv-Network performed best compared to other model used to correctly classify these audio samples generated by advanced synthetic means. Because spectrogram requires significantly more hardware and memory than processing to be able to use them, the use of Conv-Networks will most likely restrict their use to server or offline implementation as opposed to lightweight (or CPU-based) implementations.

7.2 MODELS COMPARISON

7.2.1 Comparison With ASVspoof Baseline Models

The effectiveness of the proposed method was further evaluated through a comparative assessment of the ASVspoof baseline results of established anti-spoofing models, specifically RawNet2, AASIST and LFCC-CNN. The baseline model for the ASVspoof Challenge is RawNet2, which uses a series of deep residual convolutional layers to directly process raw waveforms as input to the model during the final stages of the ASVspoof Challenge [6][7]. AASIST combines Spectro-Temporal Graph Attention (STGA) with attention mechanisms to extract features used to identify spoofed audio from real audio and was found to perform best among the established anti-spoofing algorithms from the latest ASVspoof evaluations [2].

In comparison to AASIST, LFCC-CNN, an alternative to AASIST for performing audio anti-spoofing, uses Linear Frequency Cepstral Coefficients (LFCC) calculated from audio signals through the frequency domain as input and Convolutional Neural Networks (CNN). [6,17].

While both AASIST and RawNet2 exhibit superior performance when compared to traditional benchmarking scores, the deep architecture used in these two models also requires significantly more computational resources and lower levels of GPU usage than their traditional counterparts [2][6]. However, when using Mel-frequency cepstral coefficients (MFCC) along with using spectral features when processing data for anti-spoofing purposes, traditional classifier algorithms were still able to produce competitive results through less processing overhead [9][16].

7.2.2 CNN Architecture Specification

For the CNN architecture to be reproducible. The audio files are recorded at 16 kHz and the result is extracted through 40 MFCCs (Mel Frequency Cepstral Coefficients), which represent

the audio in time-frequency using how people hear [3]. CNN is a widely used method for classifying images using their spectrograms for spoof detection [14 ,20].

The CNN consists of three convolutional blocks, with 32 filters (3x3) in the first, 64 (3x3) in the second, and 128 (3x3) in the third. Each block is followed by: Batch Normalization, ReLU activation, Max Pooling (2x2), and Dropout (0.25-0.30). The last fully connected layer has 128 neurons with ReLU activation and Dropout (0.5) to reduce overfitting. The final layer is a single Sigmoid neuron that will classify samples into two classes. The CNN is trained using the Adam optimizer and binary cross-entropy loss function [14].

Due to the structure of the CNN, it will also allow for accurate documentation, reproducibility of results, and congruency with available literature describing CNN-based spoof detection [12 ,14].

7.2.3 Comprehensive Results And Evaluation Metrics

In addition to evaluating algorithm effectiveness based on accuracy, the authors of the paper used additional evaluation criteria, including precision, recall, F1 score, and ROC curve area under the curve (AUC), as referenced by other researchers in the field of detection and prevention of spoofing attacks [19]. The implemented models were evaluated against standard classification metrics, including accuracy (ACC), precision (P), recall (R), F1 (F1), and area under the ROC curve (AUC). Table 2 summarises these results.

Model	**Acc. (%)**	**Precision**	**Recall**	**F1-Score**	**ROC-AUC**
Random Forest (RF)	**97.2**	0.97	0.98	**0.975**	**0.99**
Support Vector Machine	94.3	0.94	0.93	0.935	0.96
Convolutional Neural Network	88.6	0.89	0.87	0.88	0.92

Table 3 Comparative Performance of Models

The results presented in Table 2 clearly indicate the effectiveness of the proposed feature extraction and classification approach. The high accuracy achieved by the Random Forest model demonstrates that carefully engineered features combined with efficient algorithms can outperform more complex models in certain scenarios.

Additionally, the consistency observed across different evaluation metrics suggests that the model is not biased towards any particular class. This balance is essential in applications where both false positives and false negatives can have serious consequences, such as financial fraud detection or identity verification systems.

Result Interpretation

- The Random Forest model achieved the highest performance across all metrics, indicating excellent classification capability and strong class separation.
- The Support Vector Machine also demonstrated reliable performance with balanced precision and recall.
- The Convolutional Neural Network showed comparatively lower accuracy but remains effective in detecting complex patterns in synthetic audio.

The ROC analysis showed the potential for excellent class separation and is exceptional when it comes to the performance of the Random Forest Classifier. The additional metrics (i.e., precision and recall), provide insights into one's capabilities of producing false positives or false negatives. The capability of identifying these metrics in anti-spoofing systems is essential as the inability to identify spoofed systems creates vulnerabilities for a specific authentication solution [18,19].

However, while deep Learning approaches (such as RawNet2 and AASIST) may be capable of producing a greater Level of Accuracy than the proposed model [2,6] the balanced view of the Random Forest in conjunction with accuracy, explainability of an Individual Prediction, and Computational Performance, allows for a potential real-time authentication.

Overall, the analysis confirms that the proposed system is capable of delivering reliable and consistent performance across multiple evaluation criteria. The combination of feature engineering and model selection plays a crucial role in achieving this outcome. While there is still scope for improvement, particularly in handling highly sophisticated deepfake audio, the current results establish a strong foundation for further research and development in this domain.

CHAPTER 8

CONCLUSION, LIMITATION AND FUTURE SCOPE

8.1 CONCLUSION

A machine learning driven detection system for deepfake voices was suggested in this research work to enable the classification of real human voices versus artificial/synthetic voice. The results indicated that existing traditional voice authentication technologies based on speaker identity, cannot identify traditional voice authentication based upon the emergence of advanced artificial voice generation techniques such as those being used today.

To address the above described challenge, additional relevant features used to make this determination were extracted using a number of dimensionality reduction techniques such as Mel Frequency Cepstral Coefficients (MFCCs), chroma features and spectral contrast features. When combined these features capture many of the inherent characteristics of sound and measure differences associated with synthetic speech.

Using the above defined extracted features three different machine learning methods (Random Forest, Support Vector Machine (SVM) and Convolutional Neural Network (CNN)) were trained and subsequently evaluated.

The experimental results obtained then show that the Random Forest model was more accurate than the other two methods used for this application and because the Random Forest model has low computational complexity, it will work well in a practical setting and/or a real time scenario with run on CPU-based computers. In addition the SVM model had an acceptable performance but required more computational resources than the Random Forest model but still acceptable performance in a practical setting. The CNN model showed a very good capability in detecting small patterns and identifying differences in synthetic audio, but the requirement for large datasets and large computational resources limited their practical applications.

To sum up, the system proposed is a practical solution that will allow the detection of deepfake voices using a method that has demonstrated reliable performance, cost efficiency, and scalability. This new technology could function as an added layer of security for current voice biometric authentication systems. As such, the potential effects of this technology could be significant in reducing the risk of fraud and misuse of AI-generated voices and will increase the overall security of voice-based communications systems.

As the number of voice-based communications systems continues to increase throughout the world, trust in digital communications systems will also become increasingly important. In other words, the ability to verify the authenticity of audio files has become a compulsory component of any voice-based communication system. Therefore, the capability of having the ability to automatically determine whether the voice is coming from an individual or has been generated by an artificial intelligence system, is a critical addition to the reliability of all modern communications systems.

Moreover, this investigation shows that real-world problems do not necessarily have to be complicated to find a solution. Rather, the correct choice of pertinent features along with the proper selection of machine learning models can produce highly accurate results with very little

computing resources consumed. This balance between performance and practicality makes the system a viable candidate for implementation within low resource environments

.

Furthermore, a major conclusion drawn from this research is how adaptable the proposed framework has been. Due to its modular structure, this framework can easily accommodate future advances with respect to improvements in both feature extraction and in model development. Therefore, as new technologies are developed they will be seamlessly integrated into the framework without having to completely redesign the framework itself, allowing the framework to remain effective and applicable over a long period of time.

8.2 LIMITATIONS

Although the proposed system shows promise with respect to performance, numerous limitations exist. The most obvious limitation to the current system is that it is reliant upon a pre-defined method of extracting features. While MFCCs and other spectral features have been shown to work well for many different sources of sound, these features may only capture some of the more complex changes made by new generation models of deepfake audio. As newly developed synthetic voice technologies continue to change, more advanced, adaptive methods of extracting features may be needed in order to achieve an acceptable level of detection accuracy.

The first major limitation of the proposed model is that the dataset used to train the model is fundamental to the success of the model. A smaller or less diverse dataset will likely limit the model's ability to generalize for all audio examples and result in reduced ability to detect unseen types of deepfake audio. Due to the fact that deepfake generation techniques are constantly evolving, it is likely that the model will not accurately find all of the newest methods or all of the previously unseen methods of spoofing.

When considering the deployment of such systems in the real world it is just as important to keep in mind the challenges associated with deploying them. There are many different types of recording devices, variations in the way that networks compress data, and noise present during transmission of audio data that can lead to distortion of the audio signal. The more environmental factors that exist, or the more uncontrolled the environment is, will result in a lower consistency of predictions when the system is used outside of controlled conditions.

The second major challenge with a CNN-based approach is that it requires a much greater amount of data and processing power compared to traditional methods of classification using machine learning. To train deep learning architectures, especially if advanced, requires significantly more processing power and memory than may be available in some environments.

The presence of noise, such as ambient noise in a recording or lower-quality recordings, may also negatively affect how well the system functions. These types of issues can decrease the accuracy of detection and increase the chances of misclassification.

Another limitation to the current system is that it does not have the ability to learn continuously or adaptively. The training dataset that is used for training the model is static and not updated continuously, therefore new types of deepfake audio cannot be detected by the model until it has been retrained using the new data. Due to the rapid change in attack techniques and development of new voice synthesis technologies, the current system is unable to respond quickly to new threats.

At present the system is performing binary classification (genuine or not genuine) so it will not be able to distinguish the different varieties of synthetic audio produced; for example it cannot tell apart text-to-speech, voice cloning, or replay attacks which means that its usefulness and ability to provide insight into the various forms of spoofing attacks would therefore be limited.

8.3 FUTURE SCOPE

Future directions could include making the proposed solution more highly robust and scalable. Apart from improving the quality and quantity of datasets available for training, it may also be feasible to produce a system capable of analyzing live audio streams in real-time with instant reporting back to the user thus extending usability to many practical applications including but not limited to: live authentication, fraud detection, and voice-controlled access.

In addition to this an important future direction would be the creation of a high-quality and diverse dataset which includes multiple languages, accents and deepfake audio samples. This would improve the predictive capabilities of machine learning algorithms enabling them to detect a greater range of spoof uses..

Another exciting area of exploration includes hybrid systems that combine the power of signal processing techniques with the power of deep learning architecture, thus allowing the best of both worlds through traditional features providing high levels of interpretation while complex pattern detection happens at the feature level through the use of deep learning models. These two techniques can be combined to provide increased accuracy and greater resilience against the many spoofed attacks that exist today due to their ability to incorporate both traditional feature types and complex patterns.

Utilizing an adversarial training strategy can further improve the system's resilience to future technology used for generating 'deepfakes' by training against the newest technology before it becomes available. Doing so will provide models with the ability to endure more complex and sophisticated types of spoofing attacks that will occur in the future.

Future studies can focus on making these systems easier for user access to and system integration of voice authentication detection systems by developing user-friendly design tools (such as intuitive interface) and easy-to-use APIs for integration with already existing applications; this should assist voice authentication technology in becoming more widely accessible across multiple industries including finance (banking), health care (hospitals), and cybersecurity (internet security).

Lastly, deploying the system as a cloud-based service in conjunction with a mobile/web application enables it to be used in the real world for applications such as voice authentication via telephone (banking), telephone customer service (call centres), and voice recognition (virtual assistants). A hybrid approach that combines classical machine-learning and deep-learning techniques to build a 'hybrid model' can advance a highly reliable (trustworthy) and scalable solution to detect deepfake audio.

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LIST OF PUBLICATIONS

The research project "A Machine Learning Based Framework for Deepfake Voice Detection" was accepted for publication at the conference 'International Conference on Computational Intelligence and Information Retrieval (CIIR) at Sister Nivedita University, India. This is the first and only publication to make a new contribution to Artificial Intelligence, Audio Forensics, and Cybersecurity, based on the Author's PhD research.

In order to detect Deep Fake voice samples, the Authors have developed a Hybrid model which consists of both M/L and D/L Technologies – Three Types of Machine Learning Techniques Used: Random Forest, Support Vector Machine (SVM), Convolutional Neural Networks (CNN). The Authors also used advanced techniques to extract features from speech Signals, which include Mel-Frequency Cepstral Coefficients (MFCC), Chroma Features, and Spectral Properties. The Objective of this system is to provide Reliable and Efficient Real-time Detection of Deep Fake Voice Relative to Authentic Human Voice.

This research adds to new areas of study like: deepfake identification; voice identifier safety; and Artificial Intelligence (AI) and authentication systems that are based on AI. Practical examples can be seen in; bank security; computer communications; and voice authenticated systems where voice authenticity must be established before using the system.

CONTRIBUTION OF THE PROJECT

The present project contributes significantly to the areas of audio detection systems and audio authentication systems by providing an efficient and practical audio detection framework based upon identifying synthetic speech. The primary contribution of this work is a hybrid detection model that fuses traditional machine learning algorithms with deep learning algorithms. By combining Random Forests and Support Vector Machines (SVM) through the use of Convolutional Neural Networks (CNN), the proposed system provides high detection accuracy without sacrificing computational efficiency, thus allowing for real-time use.

Another major contribution is in the area of feature engineering. The proposed system extracts and utilizes various types of acoustic features (i.e., MFCC, chroma, and spectral features) that capture minute differences between the characteristics of authentic speech produced by humans versus that produced by artificial intelligence (AI) generating devices. As a result, these features increase the models capability to differentiate between genuine speech and synthetic audio signals.

The project also demonstrates strong performance across different models:

- Random Forest achieved an accuracy of 97.2%
- SVM achieved 94.3% accuracy
- CNN achieved 88.6% accuracy

While being very accurate, the system is also sufficiently lightweight and efficient that it can run on CPU-based computers without the need for expensive high-end equipment. Therefore, it is an excellent option for deployment. The system was designed for real-time detection, and could serve as an additional security measure for existing voice authentication systems. The impact of using this technology will be felt across multiple industries, such as banking, where it could provide security against identity theft, fraud, and unauthorized access.

The project provides a balanced data set of 11,778 sample instances with an equal number of authentic audio and fraudulent audio samples. This will result in less bias when training and evaluating the performance of the system. The system architecture is designed to be modular, scalable, and flexible, resulting in a system where future enhancements and new technologies can be easily integrated.

Therefore, this research presents a low-cost and high accuracy and scalable solution to the problem of detecting AI-generated voice fraud, addressing a major issue in society, enhancing the security of voice-based technology, and supporting further research into deep fake and audio forensic technologies.